ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 0	She returns to the starting point.
2	**(d)** Newton's first law	Law of inertia.
3	**(b)** a force acts on a body and the body is displaced in the direction of the force	W = F x s.
4	**(a)** vibrating bodies	Vibration is the source.
5	**(a)** acceleration	Change in velocity per time.
6	**(c)** are equal in magnitude, opposite in direction and act on different bodies	Third law.
7	**(a)** kinetic energy	W = change in KE.
8	**(a)** amplitude	Larger amplitude, louder.
9	**(a)** velocity, acceleration and displacement	Third equation.
10	**(a)** one-sixth	g_moon is g/6.
11	**(a)** makes work easier by changing the magnitude or direction of force	Machines help us.
12	**(d)** 17.2 m	v t/2 = 344 x 0.1/2.
13	**(d)** greater than or equal to the magnitude of displacement	Path length is at least the straight-line gap.
14	**(d)** mass	More mass, more inertia.
15	**(c)** joule	1 J = 1 N m.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Distance: total path length, scalar (1); Displacement: shortest straight-line gap with direction, vector (1).	1+1
17	A body remains at rest or in uniform motion unless acted on by an external force (1); Inertia is the tendency of a body to resist a change in its state (1).	1+1
18	Product of force and displacement in the direction of force (1); Joule (1).	1+1
19	By vibrating objects (1); A material medium (solid, liquid or gas) (1).	1+1
20	Rate of change of velocity (1); m/s squared (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Distance = 3 + 4 = 7 m (1); Displacement = sqrt(3 squared + 4 squared) (1); = 5 m (1).	1+1+1
22	Net force = 5 N east (1); a = 5/2 (1); = 2.5 m/s squared (1).	1+1+1
23	Lifting: force and displacement same direction, positive (1); Lowering: force upward, displacement downward, negative (1); W = mgh = 60 J (1).	1+1+1
24	v = f lambda = 340 m/s (1); T = 1/200 (1); = 0.005 s (1).	1+1+1
	OR: Graph is a horizontal line (1); Area = 6 x 10 (1); = 60 m (1).	1+1+1
25	a = 3 m/s squared (1); v = at = 9 m/s (1); Starts from rest (1).	1+1+1
	OR: Energy = 0.1 kW x 10 h = 1 kWh (1); 1 unit (1); Cost = 6 rupees (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	v = u + at from slope (1); s = area of trapezium = ut + (1/2)at squared (1); v squared = u squared + 2as by eliminating t (1); Meaning of symbols (1); Conditions: uniform acceleration (1).	1+1+1+1+1

	OR: First law with example of a coin on a card (1); Second law F = ma with a pushed trolley (1); Third law with rocket (1); Statements correct (1); Daily-life relevance (1).	1+1+1+1+1
27	W = F s cos(theta) (1); Along: positive (1); Opposite: negative (1); Perpendicular: zero (1); Example 60 J (1).	1+1+1+1+1
	OR: Vibrating source (1); Compression and rarefaction (1); Amplitude and wavelength (1); Frequency and time period f = 1/T (1); Wave equation v = f lambda (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** a vacuum	A material medium is required.
2	**(a)** displacement	Product of v and t.
3	**(a)** act on different bodies	Different bodies.
4	**(a)** can neither be created nor destroyed but only changed from one form to another	Law.
5	**(b)** frequency	Higher frequency, higher pitch.
6	**(a)** 10 m/s squared	g about 9.8.
7	**(b)** increasing the stopping time and reducing the force	F = change in momentum/time.
8	**(d)** load / effort	MA.
9	**(a)** echolocation using ultrasonic waves	Reflected ultrasound.
10	**(b)** vector quantity	It has magnitude and direction.
11	**(d)** newton	N = kg m/s squared.
12	**(d)** zero	Force is perpendicular to displacement.
13	**(a)** longitudinal waves	Particles vibrate along the direction of travel.
14	**(d)** at rest	Distance does not change.
15	**(b)** the exhaust gases pushed downward exert an equal and opposite force on it	Third law.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Inertia of motion (1); Body tends to continue to move with the same speed (1).	1+1
17	Work done on a body equals the change in its kinetic energy (2).	2
18	Pitch depends on frequency (1); Loudness depends on amplitude (1).	1+1
19	Direction of velocity changes continuously (1); Change in velocity means acceleration (1).	1+1
20	To increase the time to stop the ball (1); This reduces the force on the hands (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Pulse sent and its echo is received (1); Depth = v t/2 (1); = 1500 x 4/2 = 3000 m (1).	1+1+1
22	Distance = 50 km in 1 h (1); Average speed 50 km/h (1); Speedometer shows instantaneous speed (1).	1+1+1
23	Foot pushes the ground backward (1); Friction pushes forward (1); On ice friction is low and we slip (1).	1+1+1
24	Height = 2.4 m (1); Work = 960 J (1); P = 120 W (1).	1+1+1
	OR: Speed increases with temperature (1); Particles move faster and transfer disturbance quickly (1); About 0.6 m/s per degree Celsius (1).	1+1+1
25	v = at = 12 m/s (1); s = (1/2)at squared = 0.5 x 3 x 16 (1); = 24 m (1).	1+1+1
	OR: a = -4 m/s squared (1); F = ma (1); = 4000 N opposite to motion (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Velocity along the tangent (1); Direction changes: accelerated (1); v = 2 x 3.14 x 0.5/2 (1); = 1.57 m/s (1); Speed constant (1).	1+1+1+1+1
	OR: Net force 20 N (1); a = 4 m/s squared (1); v = 16 m/s (1); s = 32 m (1); Formulas used (1).	1+1+1+1+1

27	At the extreme: PE max, KE zero (1); At the mean: KE max, PE zero (1); Total constant (1); Air resistance reduces amplitude (1); Energy converted to heat (1).	1+1+1+1+1
	OR: Repeated reflections (1); Persists and blurs speech (1); Absorbing materials: curtains, carpets (1); Curved sound boards reflect sound to the audience (1); Roof design (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** rate of doing work	P = W/t.
2	**(a)** distinguish two sounds of the same pitch and loudness	Waveform differs.
3	**(d)** 20 m/s	v = gt.
4	**(a)** increase the time of stopping and reduce the force	Impulse.
5	**(d)** fulcrum	Pivot.
6	**(b)** sound navigation and ranging	Uses ultrasonic waves.
7	**(d)** scalar quantity	Only magnitude.
8	**(b)** 1 kg	F = ma.
9	**(d)** 50 J	F s.
10	**(c)** compression	High pressure region.
11	**(b)** uniform velocity	Velocity constant.
12	**(d)** opposite to the direction of motion or tendency of motion	Opposes relative motion.
13	**(b)** watt	1 W = 1 J/s.
14	**(c)** 20 Hz to 20000 Hz	Normal hearing.
15	**(a)** zero	Instantaneously at rest.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Changes the direction of the force (1); Convenient to pull downward (1).	1+1
17	20 Hz to 20 kHz (1); Below 20 Hz and above 20 kHz respectively (1).	1+1
18	Speed: distance/time, scalar (1); Velocity: displacement/time, vector (1).	1+1
19	Force is the rate of change of momentum (1); F = ma (1).	1+1
20	When force is opposite to displacement (1); Friction on a sliding block (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Work to accelerate from rest: W = F s (1); F = ma, v squared = 2as (1); KE = (1/2) m v squared (1).	1+1+1
22	lambda1 = 344/20 = 17.2 m (1); lambda2 = 344/20000 (1); = 0.0172 m = 1.72 cm (1).	1+1+1
23	a = 20/10 = 2 m/s squared (1); s = (10 + 30)/2 x 10 (1); = 200 m (1).	1+1+1
24	Action and reaction act on different bodies (1); Horse pushes the ground backward, ground pushes it forward (1); Net force on the horse-cart system comes from friction of the ground (1).	1+1+1
	OR: l = sqrt(30 squared + 40 squared) = 50 cm (1); MA = l/h (1); = 50/30 = 1.67 (1).	1+1+1
25	Pitch: frequency (1); Loudness: amplitude (1); Quality/timbre: waveform (1).	1+1+1
	OR: v = u - gt gives 0 = 20 - 10t, t = 2 s (1); h = u squared / 2g = 400/20 (1); = 20 m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Stage 1: a = 5 m/s squared (1); Stage 2: a = 0 (1); Stage 3: a = -6 m/s squared (1); Distance: 90 + 300 + 75 (1); = 465 m (1).	1+1+1+1+1

	OR: Place a card on a glass with a coin on it (1); Flick the card quickly (1); The coin drops into the glass (1); Pile of coins: strike the bottom coin (1); Inertia of rest (1).	1+1+1+1+1
27	W = F s (1); v squared - u squared = 2as (1); W = (1/2)m v squared - (1/2) m u squared (1); 100 = (1/2)(2) v squared (1); v = 10 m/s (1).	1+1+1+1+1
	OR: v = distance/time = lambda/T (1); = lambda f (1); v = 256 x 1.34 (1); = 343 m/s (1); Units (1).	1+1+1+1+1

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